

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

QUESTIONS: 30

Time Duration: 30 Minutes

Total Marks:30

Annexure A

MCQ

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Circle the most appropriate option for each question.

Q1. Which of the following best describes a Logical Agent in AI?

- a) An agent that reacts randomly to the environment
- b) An agent that uses a knowledge base and logical inference to decide actions
- c) An agent that only follows pre-defined scripts
- d) An agent that learns from rewards

Q2. In Propositional Logic, ' $P \rightarrow Q$ ' is logically equivalent to:

- a) $P \wedge \neg Q$
- b) $\neg P \vee Q$
- c) $\neg Q \rightarrow \neg P$ only
- d) Both b and c

Q3. Conjunctive Normal Form (CNF) requires the formula to be expressed as:

- a) A disjunction of conjunctions
- b) A conjunction of disjunctions (clauses)
- c) A conjunction of literals only
- d) A sequence of implications

Q4. In FOL, the sentence $\forall x: \text{Student}(x) \rightarrow \text{Passes}(x)$ means:

- a) Some students pass
- b) Every entity that is a student passes
- c) If something passes, it is a student
- d) There exists at least one student who passes

Q5. Unification in FOL is the process of finding substitutions that:

- a) Convert clauses to CNF
- b) Make two logical expressions syntactically identical
- c) Prove a goal using backward chaining
- d) Derive facts using Modus Ponens

Q6. In the Resolution algorithm, deriving the empty clause indicates:

- a) The knowledge base is consistent
- b) The negated goal is satisfiable
- c) A contradiction — the original goal is proven
- d) More clauses need to be added

Q7. Forward Chaining is best described as a _____ strategy.

- a) Goal-driven / top-down
- b) Data-driven / bottom-up
- c) Heuristic-based depth-first
- d) Probabilistic inference

Q8. Backward Chaining begins reasoning from:

- a) All known facts toward a conclusion
- b) The goal and works backward to verify supporting facts
- c) A random clause in the knowledge base
- d) The resolution refutation tree

Q9. Which step is NOT part of converting a sentence to CNF?

- a) Eliminate biconditionals ($\leftrightarrow$)
- b) Move negations inward using De Morgan's laws
- c) Apply Skolemization to remove existential quantifiers
- d) Distribute AND over OR

Q10. Knowledge Engineering in FOL involves:

- a) Training a neural network on domain-specific data
- b) Defining domain concepts, relations, and axioms formally in FOL
- c) Writing heuristic search strategies
- d) Encoding reward functions for reinforcement learning

Q11. Modus Ponens states: If $P \rightarrow Q$ is known and P is true, then:

- a) $\neg Q$ is true
- b) Q must be true
- c) P is uncertain
- d) The KB is inconsistent

Q12. The Most General Unifier (MGU) is the substitution that:

- a) Is the most specific possible match
- b) Makes two terms identical with minimal variable binding
- c) Eliminates all variables from a clause
- d) Applies resolution to produce CNF

Q13. Which inference method is more appropriate when a specific goal is known?

- a) Forward Chaining
- b) Truth Table method
- c) Backward Chaining
- d) A* search

Q14. Propositional Logic differs from First Order Logic because it:

- a) Supports probabilistic reasoning
- b) Cannot represent objects, relations, or quantifiers — only truth values of atomic propositions
- c) Is always more expressive
- d) Handles uncertainty better

Q15. Resolution Refutation proves a goal G by:

- a) Directly deriving G from the KB using forward chaining
- b) Adding $\neg G$ to the KB and deriving a contradiction (empty clause)
- c) Checking G in the truth table
- d) Applying Modus Tollens repeatedly

Q16. Learning from observation in AI refers to:

- a) An agent receiving numerical rewards from the environment
- b) Improving performance by examining examples or experience from the environment
- c) Acquiring knowledge by following explicit symbolic rules
- d) Memorizing a fixed dataset without generalization

Q17. Inductive learning is best described as:

- a) Deriving specific conclusions from general rules
- b) Generalizing a hypothesis from specific training examples
- c) Memorizing training data exactly
- d) Applying search algorithms to find optimal policies

Q18. In a Decision Tree, what does each internal node represent?

- a) A class label / output decision
- b) A probability distribution over outcomes
- c) A test on an attribute / feature
- d) A training data sample

Q19. Explanation-Based Learning (EBL) differs from inductive learning because it:

- a) Requires large amounts of training data to generalize
- b) Learns only from numeric data using regression
- c) Uses prior domain knowledge to explain and generalize a single example
- d) Applies unsupervised clustering to find hidden patterns

Q20. In statistical learning, a Naive Bayes classifier assumes:

- a) All features are conditionally dependent on each other given the class
- b) The data must follow a uniform distribution
- c) All features are conditionally independent given the class label
- d) Labels are determined by a reward function

Q21. Learning with complete data (no hidden variables) in a Bayesian network involves:

- a) Using the EM algorithm to estimate missing values
- b) Training a deep neural network with dropout regularization
- c) Directly computing maximum likelihood estimates from observed data
- d) Applying reinforcement signals to update weights

Q22. The Expectation-Maximization (EM) algorithm is used when:

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| a) Training data is fully labeled and no variables are hidden | c) The dataset contains hidden (latent) variables that cannot be directly observed |
| b) An agent explores an unknown environment using trial and error | d) Backpropagation fails to converge in a neural network |

Q23. In a Neural Network, the activation function serves to:

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| a) Initialize the weights of each neuron to zero | c) Introduce non-linearity to enable learning of complex patterns |
| b) Normalize the input dataset before training | d) Store previously learned examples in memory |

Q24. Which of the following correctly describes backpropagation in a neural network?

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| a) Weights are updated in the forward pass using input gradients | c) The error is propagated backward from output to input layers to adjust weights |
| b) Outputs of each layer are stored and reused in later epochs | d) A separate model is trained for each hidden layer independently |

Q25. A Support Vector Machine (SVM) — a type of kernel machine — classifies data by:

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| a) Minimizing the number of training examples used | c) Finding the maximum-margin hyperplane that separates classes |
| b) Grouping similar data points into clusters using centroids | d) Updating weights using Q-values from an environment |

Q26. The kernel trick in kernel machines (SVMs) allows:

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| a) Reducing the number of support vectors needed | c) Operating in a high-dimensional feature space without explicitly computing coordinates |
| b) Accelerating gradient descent convergence | d) Replacing hidden layers in a deep neural network |

Q27. In Reinforcement Learning, the agent aims to maximize:

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| a) Classification accuracy on a fixed test set | c) Cumulative reward received over time through interaction with the environment |
| b) The number of features selected from the training data | d) The size of the neural network hidden layer |

Q28. The Q-Learning algorithm in Reinforcement Learning is classified as:

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| a) A supervised learning algorithm requiring labeled datasets | c) A model-free, off-policy temporal difference learning method |
| b) An unsupervised clustering technique using centroids | d) A decision-tree-based ensemble classifier |

Q29. Overfitting in machine learning occurs when:

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| a) The model performs poorly on both training and test data | c) The model fits training data too closely and generalizes poorly to new data |
| b) The learning rate is set too low during gradient descent | d) The training set has more features than training examples |

Q30. Which of the following best distinguishes Reinforcement Learning from Supervised Learning?

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| a) RL uses labeled input-output pairs; SL uses reward signals | c) RL learns via trial-and-error with reward feedback; SL learns from labeled examples |
| b) Both RL and SL require explicit reward functions | d) RL is always faster to converge than SL on any dataset |

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

